

Unit 5

Application of Flip Flops

Design of Shift Register using D-flip flop, Design of Counters: Asynchronous counters using D and T-flip flop, Synchronous Counters using D-flip flop and T Flip Flop.

Registers

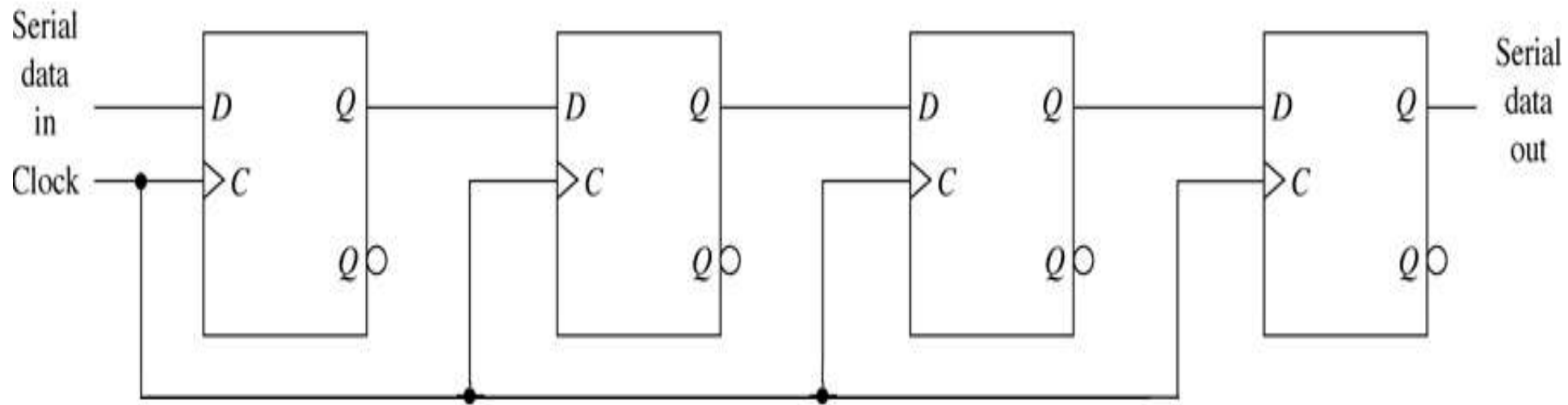
- A Collection of Flip-Flops taken as an entity.
- Basic Function is to hold the information within a digital system.
- The finite number of 0-1 combinations that are stored in the register is known as ***state or content***
- Shift Registers are capable of moving information position-wise on occurrence of clock.

Classification of Registers

- Depending on Data Movement
 - Unidirectional: Capable of moving data only on one side
 - Bidirectional: Capable of moving data both sides
- Manner in which the information is entered and outputted
 - Serial In-Serial Out [SISO]: Data entered serially and read serially
 - Serial In-Parallel Out[SIPO]: Data entered serially, read in parallel
 - Parallel In-Serial Out[PISO]: Data entry parallel, read serially
 - Parallel In-Parallel Out[PIPO]: Both entry & reading in parallel

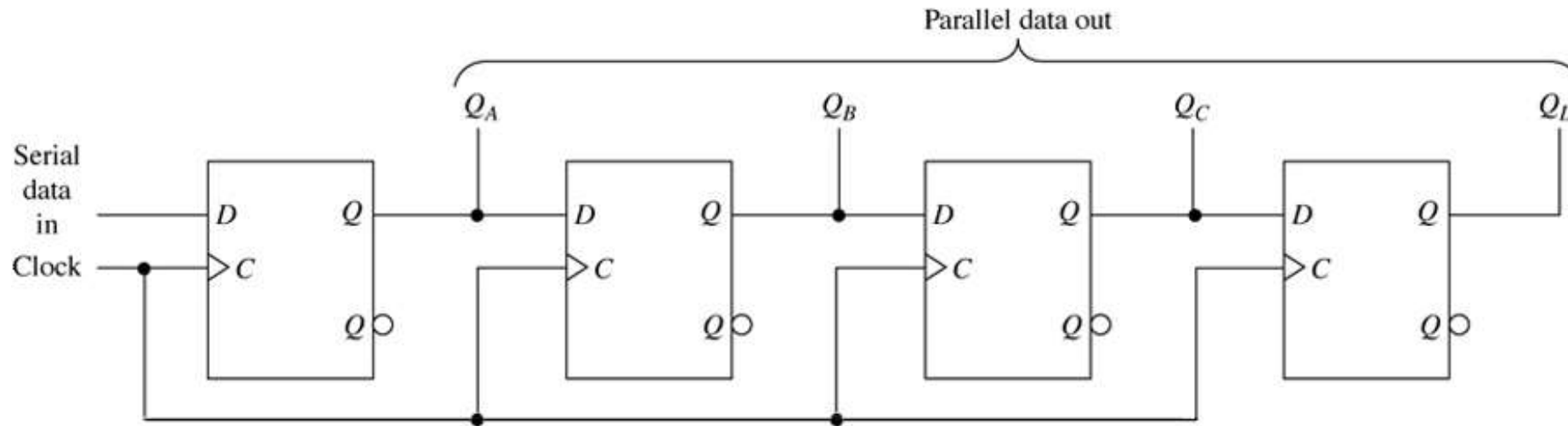
Serial In Serial Out Unidirectional Register

- Capable of shifting data to the right
- Signal Value shifted becomes available as an output after four-clock pulses

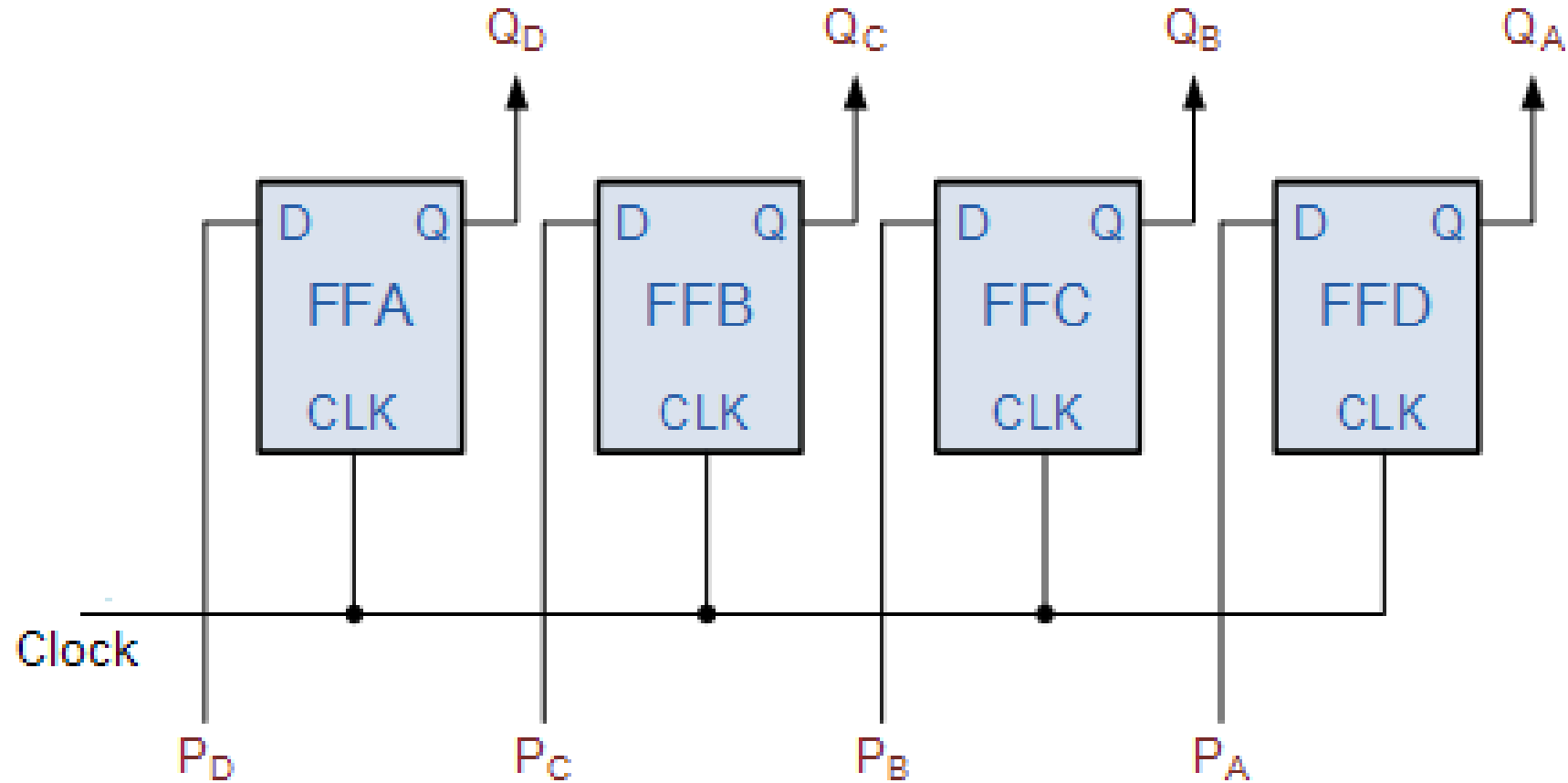


Serial In Parallel Out Unidirectional Register

- Outputs provided from each flip-flop.
- Information shifted in serially and, after appropriate number of shifts, made available in parallel.
- Can be used in Serial-to-Parallel conversion



Parallel-in Parallel-out shift register



Counters

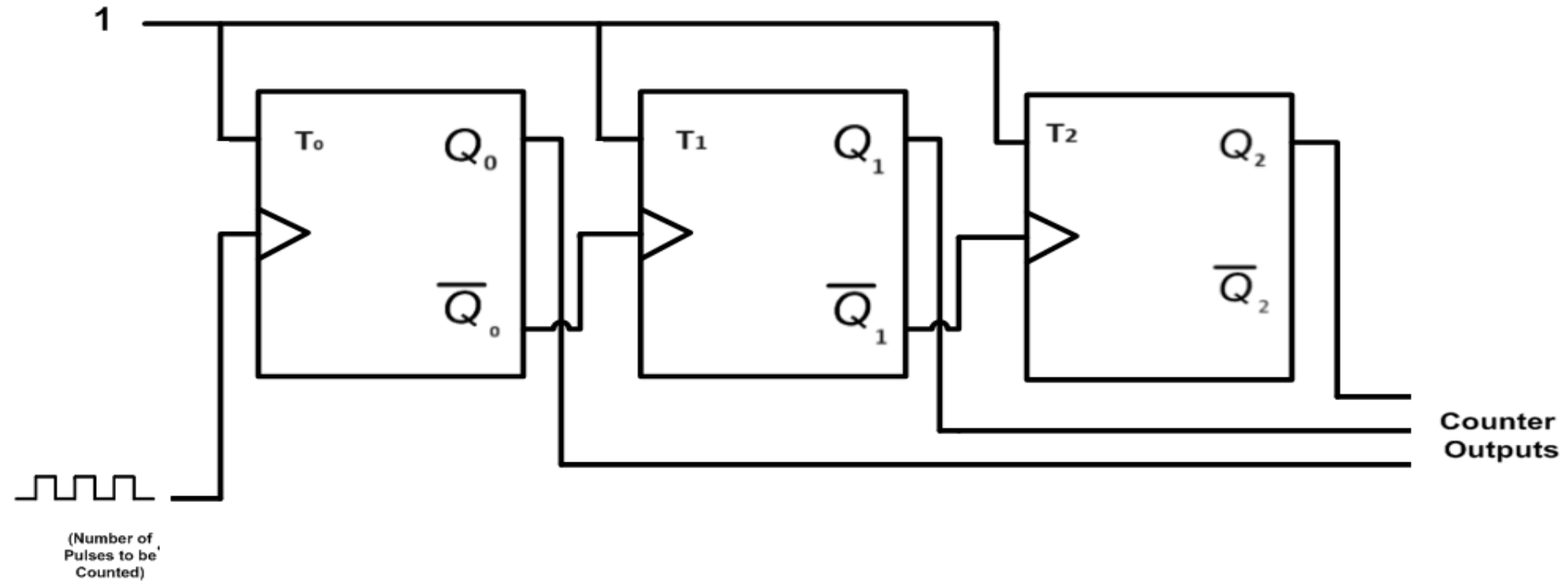
- A *counter* is a sequential circuit that goes through a predetermined sequence of states upon the application of clock pulses.
- Each output of the sequence is dependent on the contents of the flip flops and is called a state of the counter.
- The Modulus of a counter is the total number of states of the counter.

Eg: mod-9, mod-6 counter.

- Counter with m states is called a modulus m or mod- m counter
- Such counters can be represented using a state diagram

- Counters are categorized as:
 - **Asynchronous (Ripple) Counters:** The FF output transition serves as a source for triggering other FFs. No common clock.
 - **Synchronous Counter:** All FFs receive the common clock pulse, and the change of state is determined from the present state.

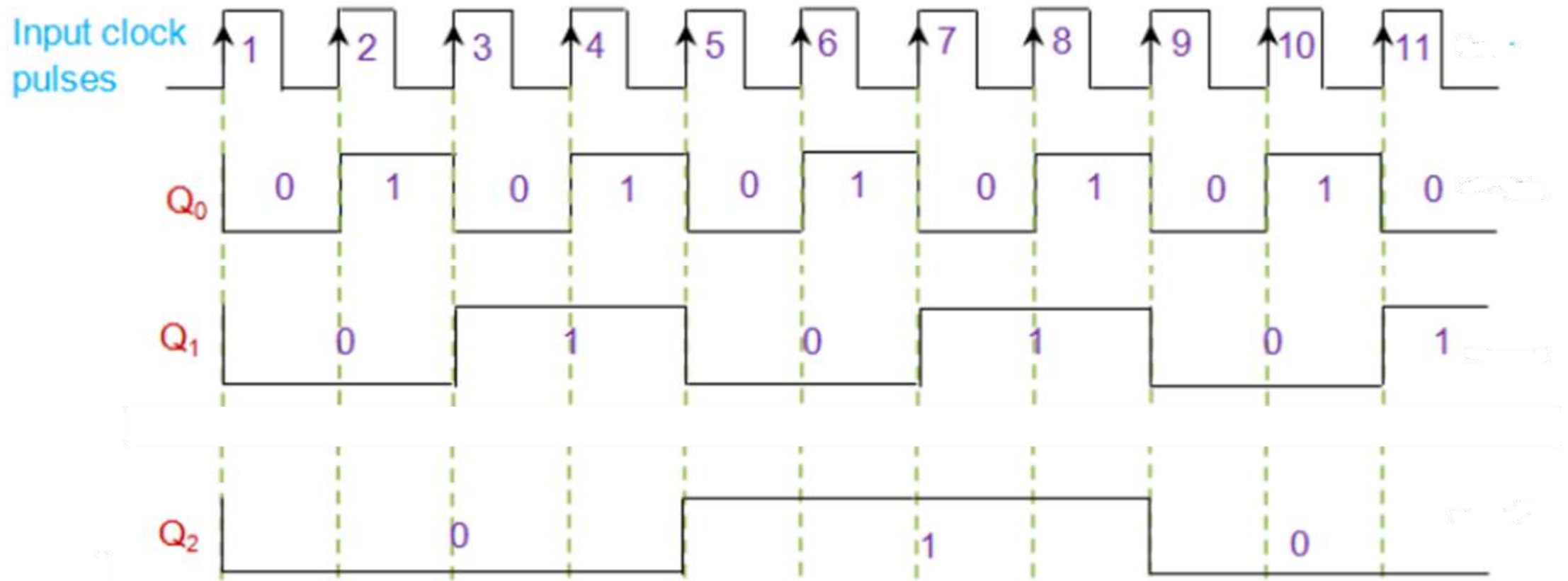
3-bit Ripple Up Counter(positive edge triggered T flip-flop)



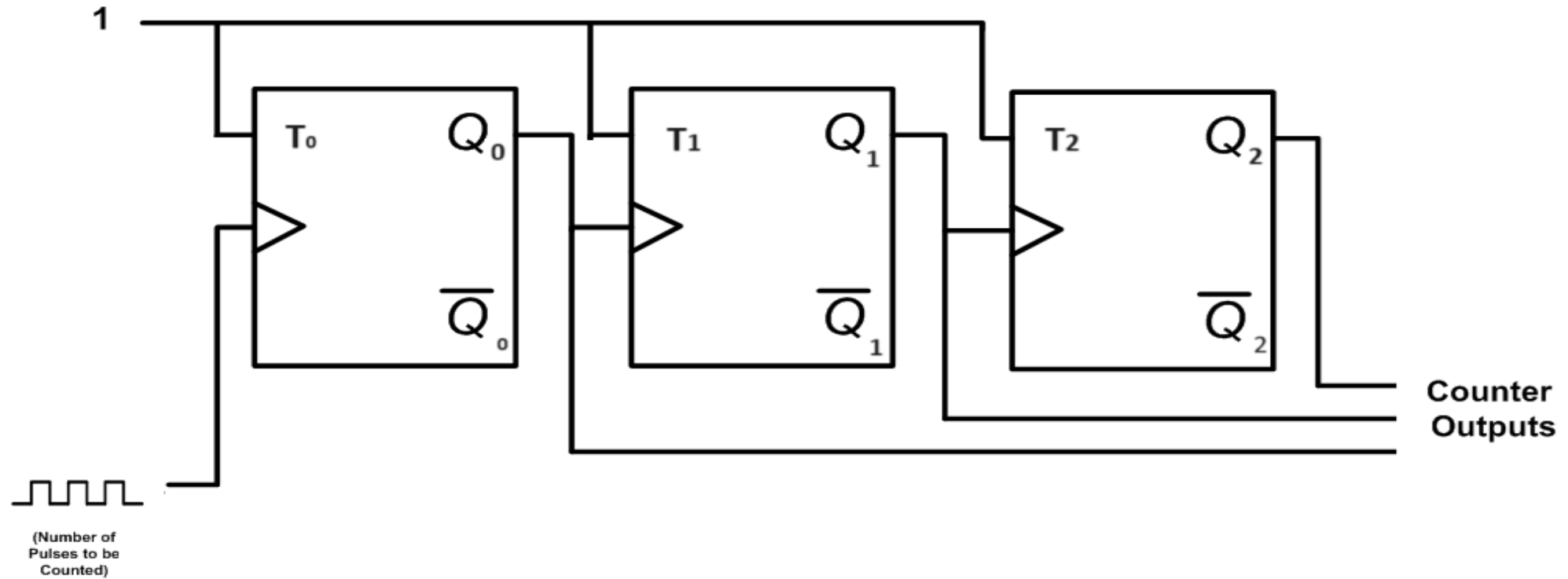
Count Sequence

Clock	Q₂ (MSB)	Q₁	Q₀ (LSB)
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Timing Diagram



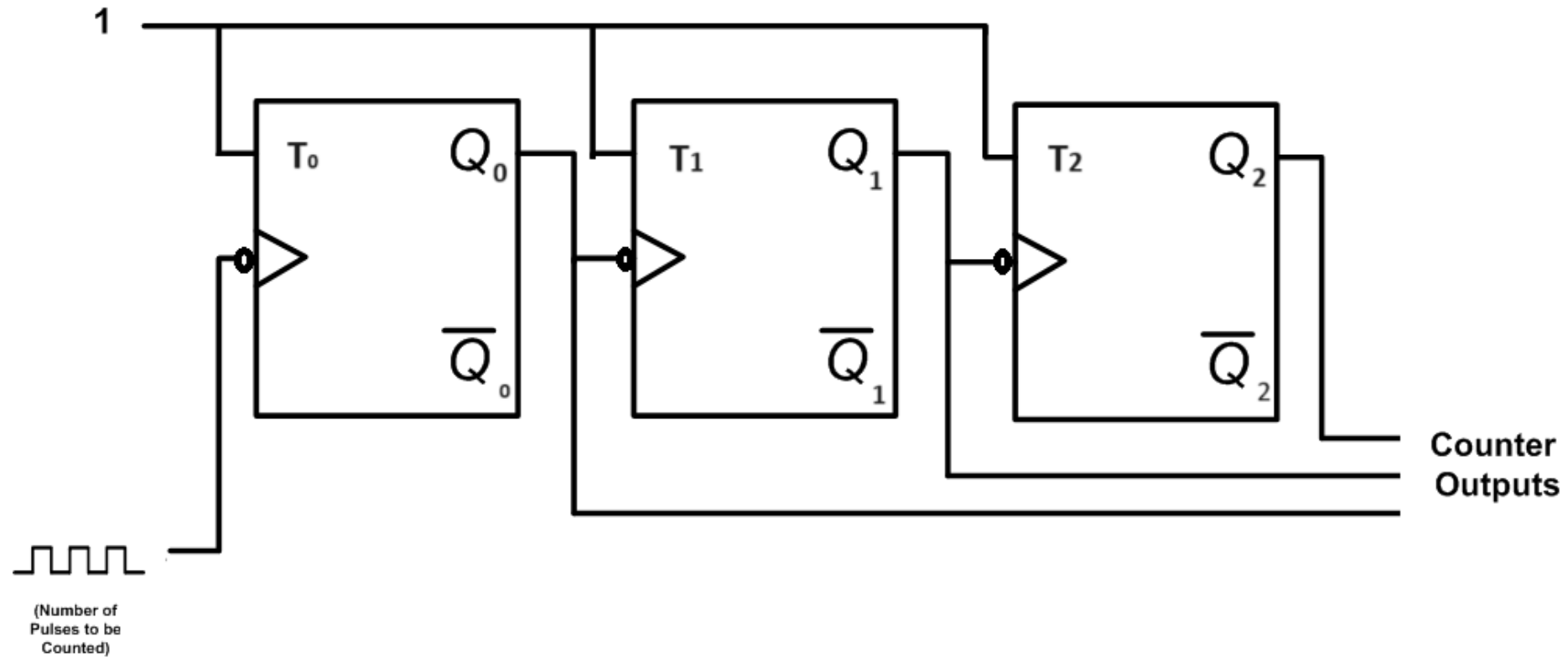
3-bit Ripple Down Counter(positive edge triggered T flip-flop)



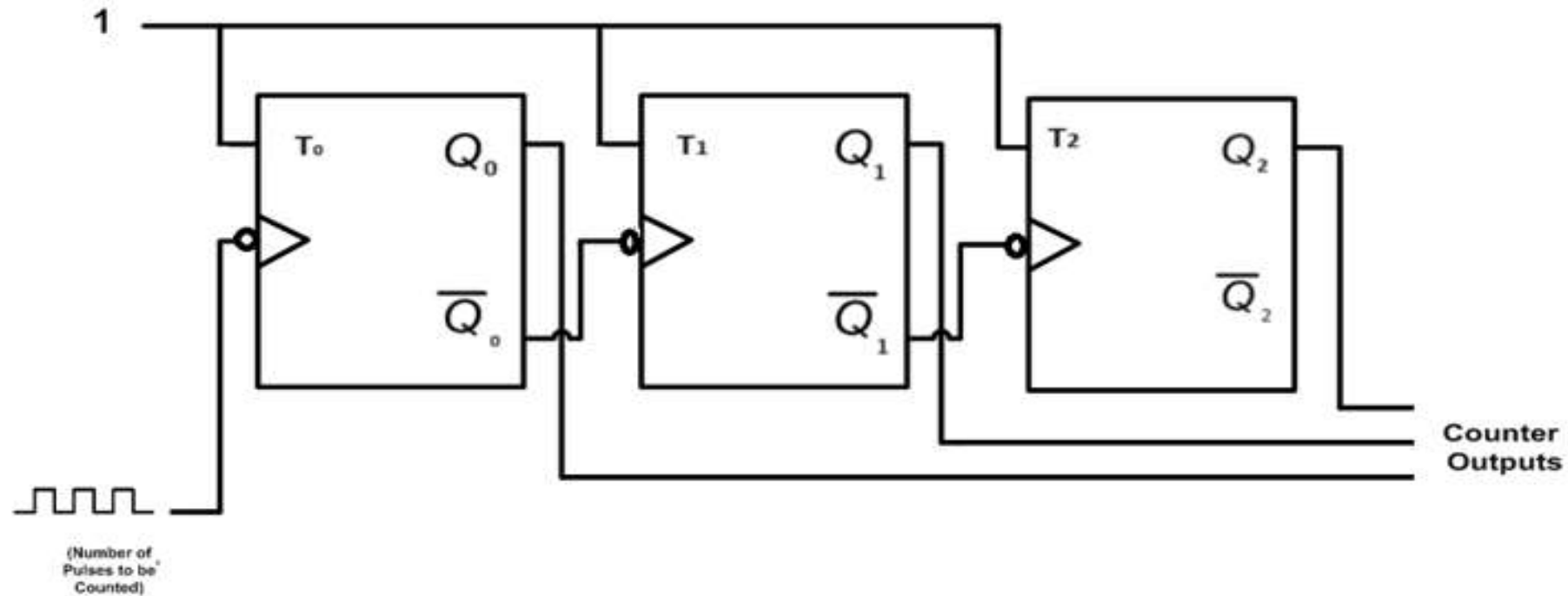
Count Sequence

Clock	Q₂ (MSB)	Q₁	Q₀ (LSB)
0	1	1	1
1	1	1	0
2	1	0	1
3	1	0	0
4	0	1	1
5	0	1	0
6	0	0	1
7	0	0	0

3-bit Ripple Up Counter(negative edge triggered T flip-flop)



3-bit Ripple Down Counter(negative edge triggered T flip-flop)

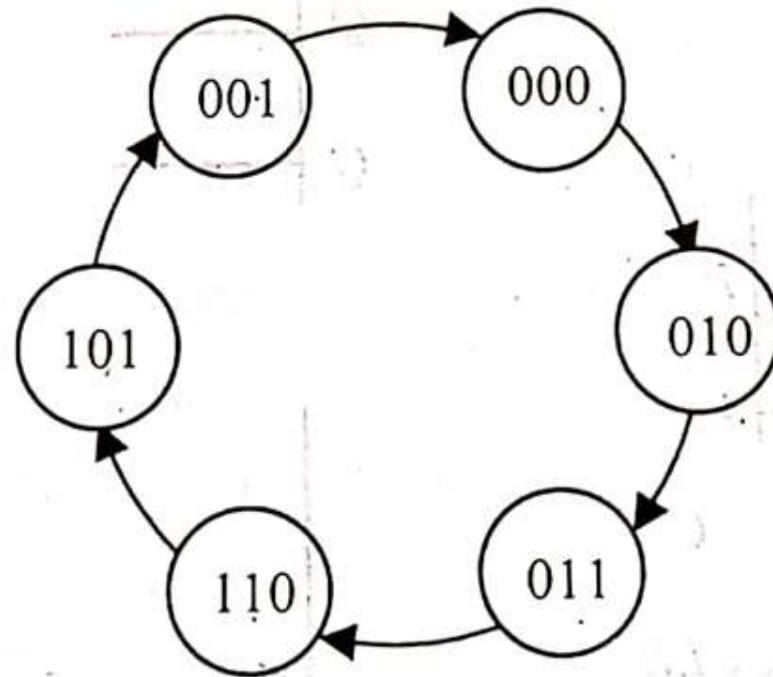


Design of a Synchronous mod-6 Counter using Clocked D Flip-Flops

Application table for a clocked D flip-flop

<i>Row no.</i>	$Q \rightarrow Q^+$		D
0	0	0	0
1	0	1	1
2	1	0	0
3	1	1	1

The state diagram for the sequence is shown in Fig.



Excitation table for mod-6 counter

Cell no.	Present state			Next state			Flip-flop Inputs		
	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1	0
1	0	0	1	0	0	0	0	0	0
2	0	1	0	0	1	1	0	1	1
3	0	1	1	1	1	0	1	1	0
4	1	0	0	—	—	—	—	—	—
5	1	0	1	0	0	1	0	0	1
6	1	1	0	1	0	1	1	0	1
7	1	1	1	—	—	—	—	—	—

	$Q_1 Q_0$			
	00	01	11	10
0	0	0	1	0
	0	1	3	2
1	-	0	-	1
	4	5	7	6

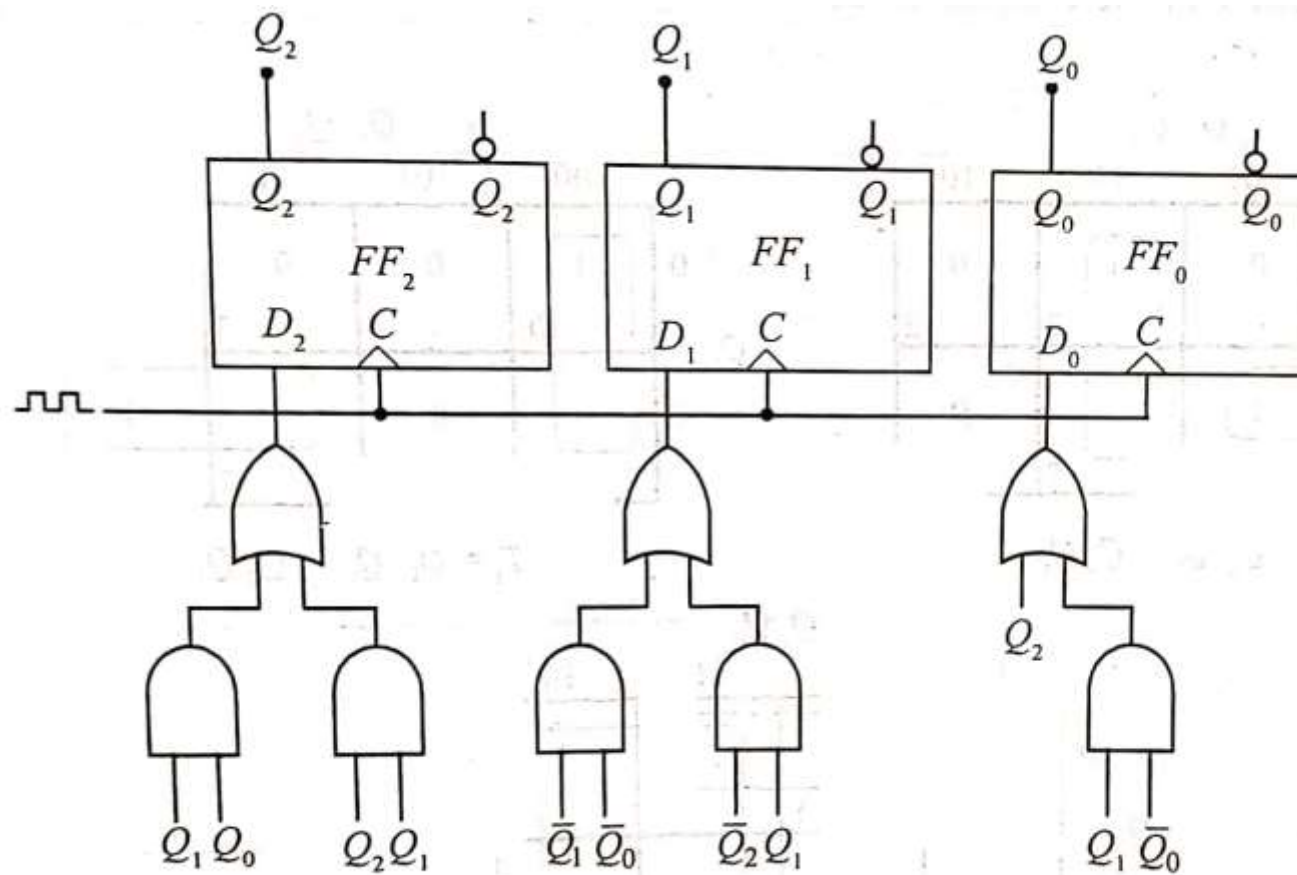
$$D_2 = Q_1 Q_0 + Q_2 Q_1$$

	$Q_1 Q_0$			
	00	01	11	10
0	1	0	1	1
	0	1	3	2
1	-	0	-	0
	4	5	7	6

$$D_1 = \bar{Q}_1 \bar{Q}_0 + \bar{Q}_2 Q_1$$

	$Q_1 Q_0$			
	00	01	11	10
0	0	0	0	1
	0	1	3	2
1	-	1	-	1
	4	5	7	6

$$D_0 = Q_2 + Q_1 \bar{Q}_0$$

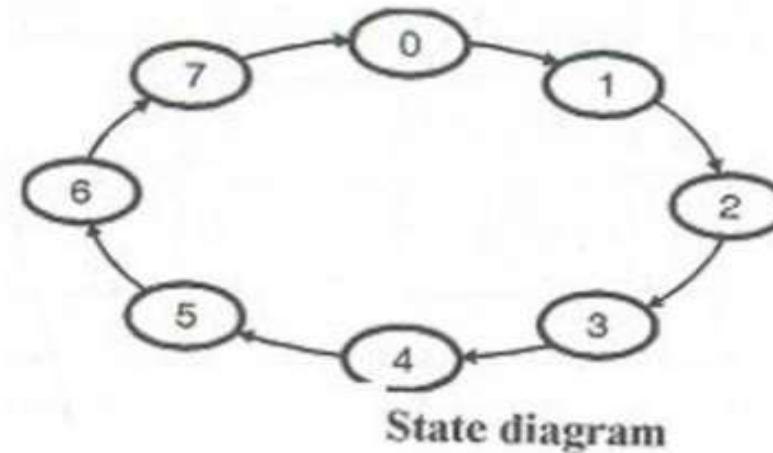


Implementation of mod-6 synchronous counter using D flip-flops

Design 3 bit synchronous counter using T F/F

Step 3 : State diagram and circuit excitation table :

Q_C	Q_B	Q_A
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1



Circuit excitation table

Present state			Next state			Flip flop input		
Q_C	Q_B	Q_A	Q_{C+1}	Q_{B+1}	Q_{A+1}	T_C	T_B	T_A
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

For T_C

	$Q_B Q_A$	00	01	11	10
Q_C	0	0	0	1	0
	1	0	0	1	0

$Q_B Q_A$

$$\therefore T_C = Q_B Q_A$$

For T_B

	$Q_B Q_A$	00	01	11	10
Q_C	0	0	1	1	0
	1	0	1	1	0

Q_A

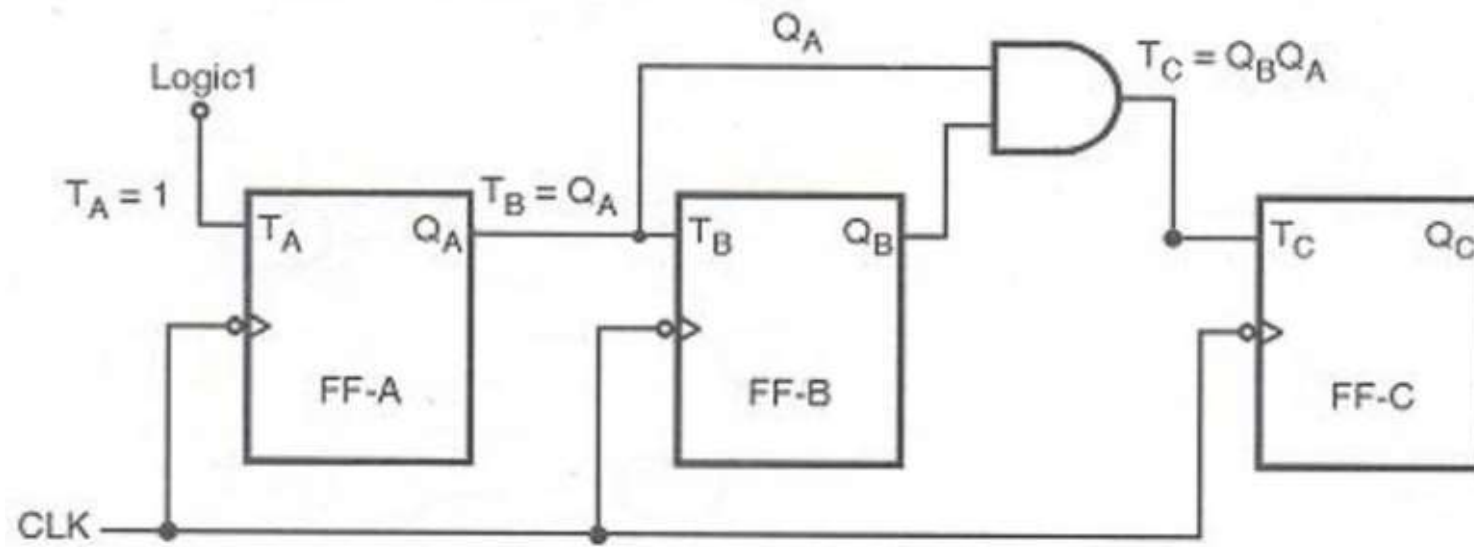
$$T_B = Q_A$$

For T_A

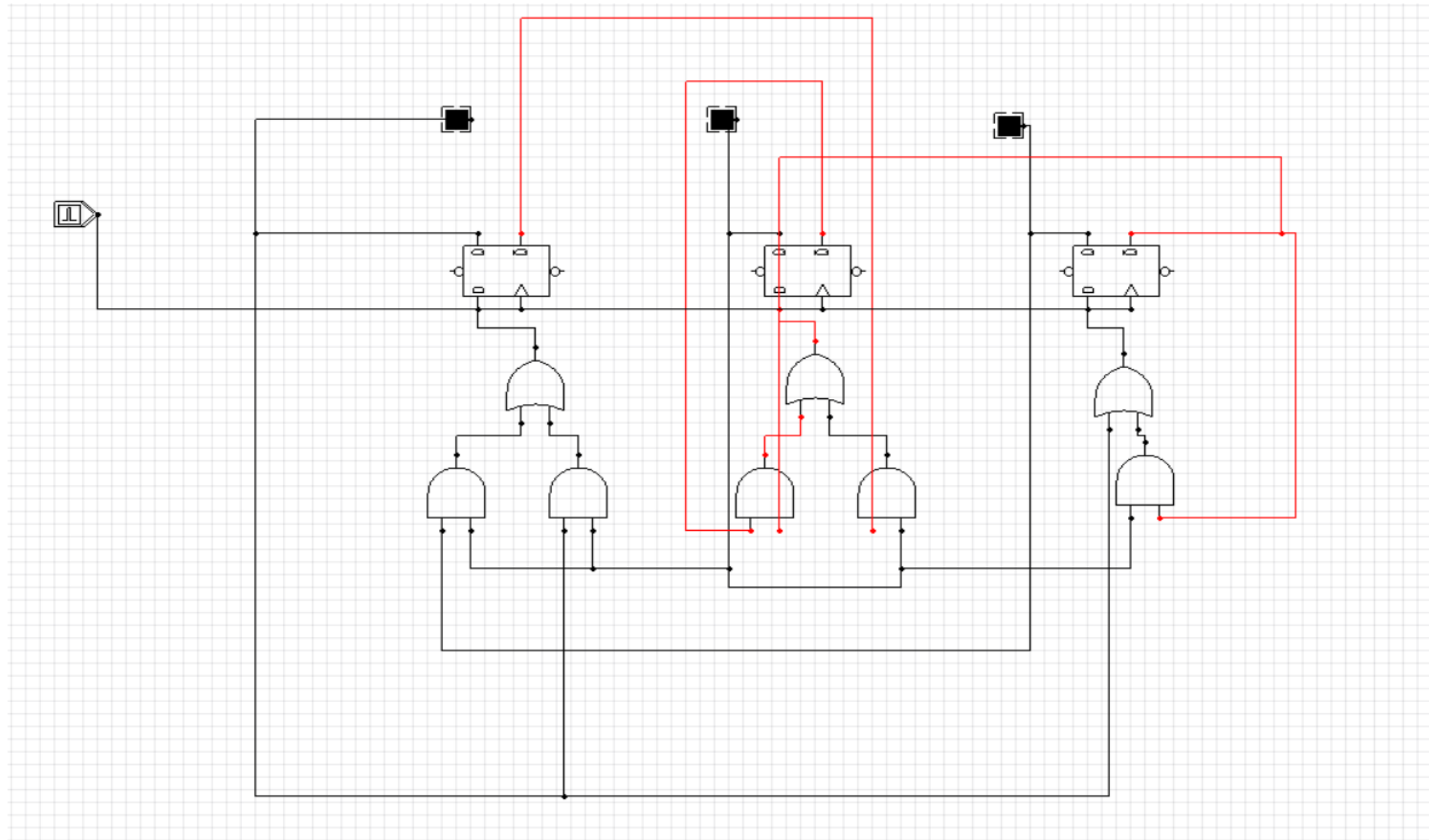
	$Q_B Q_A$	00	01	11	10
Q_C	0	1	1	1	1
	1	1	1	1	1

$T_A = 1$

$$T_A = 1$$



Logic diagram of a 3-bit synchronous counter



Design a synchronous mod-16 counter using positive edge triggered D flip-flops with minimal combinational gate circuits.

Excitation table for mod-16 counter

Cell no.	Present states				Next states				Flip-flop inputs			
	Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	1	0	0	0	1
1	0	0	0	1	0	0	1	0	0	0	1	0
2	0	0	1	0	0	0	1	1	0	0	1	1
3	0	0	1	1	0	1	0	0	0	1	0	0
4	0	1	0	0	0	1	0	1	0	1	0	1
5	0	1	0	1	0	1	1	0	0	1	1	0
6	0	1	1	0	0	1	1	1	0	1	1	1
7	0	1	1	1	1	0	0	0	1	0	0	0
8	1	0	0	0	1	0	0	1	1	0	0	1
9	1	0	0	1	1	0	1	0	1	0	1	0
10	1	0	1	0	1	0	1	1	1	0	1	1
11	1	0	1	1	1	1	0	0	1	1	0	0
12	1	1	0	0	1	1	0	1	1	1	0	1
13	1	1	0	1	1	1	1	0	1	1	1	0
14	1	1	1	0	1	1	1	1	1	1	1	1
15	1	1	1	1	0	0	0	0	0	0	0	0

5. 0. 0. 0.

		$Q_1 Q_0$			
		00	01	11	10
$Q_3 Q_2$	00	0	0	0	0
		0	1	3	2
	01	0	0	①	0
		4	5	7	6
11		1	1	0	1
		12	13	15	14
10		1	1	1	1
		8	9	11	10

$$D_3 = \bar{Q}_3 Q_2 Q_1 Q_0 + Q_3 \bar{Q}_2 + Q_3 \bar{Q}_1 + Q_3 \bar{Q}_0$$

		$Q_1 Q_0$			
		00	01	11	10
$Q_3 Q_2$	00	0	0	1	0
		0	1	3	2
	01	1	1	0	1
		4	5	7	6
11		1	1	0	1
		12	13	15	14
10		0	0	1	0
		8	9	11	10

$$D_2 = \bar{Q}_2 Q_1 Q_0 + Q_2 \bar{Q}_1 + Q_2 \bar{Q}_0$$

	$Q_1 Q_0$				
	00	01	11	10	
$Q_3 Q_2$	00	0 0	1 1	0 3	1 2
	01	0 4	1 5	0 7	1 6
	11	0 12	1 13	0 15	1 14
	10	0 8	1 9	0 11	1 10

$$D_1 = \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0 = Q_1 \oplus Q_0$$

	$Q_1 Q_0$			
	00	01	11	10
$Q_3 Q_2$	00	1 0	0 1	0 3 2
	01	1 4	0 5	0 7 6
	11	1 12	0 13	0 15 14
	10	1 8	0 9	0 11 10

$$D_0 = \bar{Q}_0$$